

Report No.:

Test Time: 2024/10/15 11:37:18

LUMINAIRE PHOTOMETRIC TEST REPORT

Luminaire Property

Manufacturer: 0

Luminaire Type: AL9333 4000K 80RA

Lamp Type:

Luminous Width: 0 mm

Luminous Length: 0 mm

Luminous Height: 0 mm

Voltage(V): 219.30 V

Current(A): 0.0890 A

Power(W): 19.13 W

Power Factor: 0.9720

Lamp Luminous Flux: 1461.58

Photometric Results

CIE Class: Direct

Luminaire Luminous Flux: 1461.575 lm

Downward Ratio: 100.00%

Luminaire Efficacy Rating(LER): 76.40 lm/W

Beam Angle(50%) C0-180,C90-270: 84.1,84.0

Field Angle(10%) C0-180,C90-270: 132.2,133.0

S/MH C0-180,C90-270: 1.109,1.109

Max. Intensity: 774.65 cd

Max.Intensity Angle: C=180 Gamma=1

Upward Ratio: 0.00%

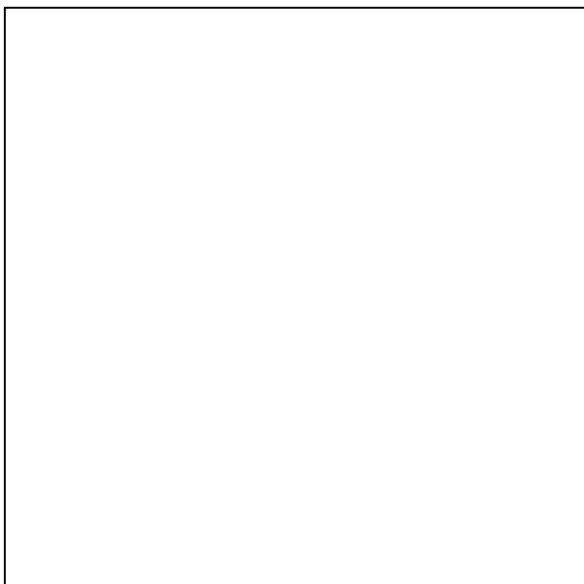
Efficacy: 100.00%

Center Intensity: 774.10 cd

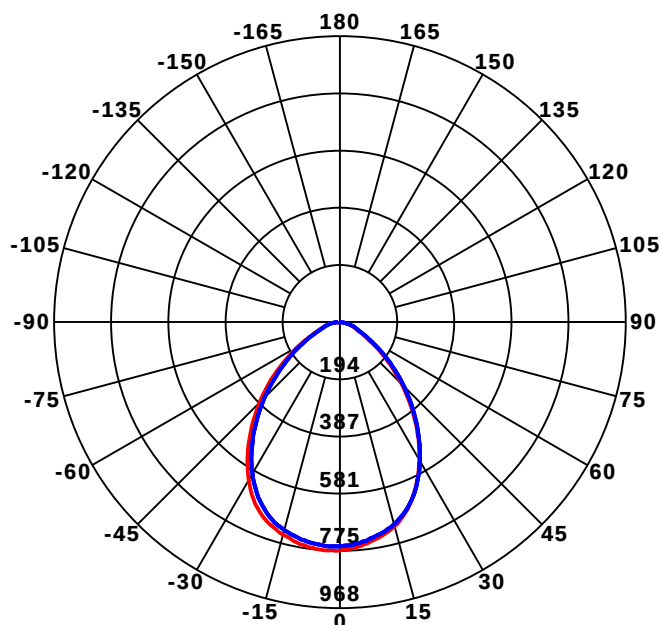
Effective Luminous Flux: 1378.82 lm

ERP(ϕ use)(120deg): 1310.33 lm

Picture of Luminaire



Luminous Intensity Distribution Curve



Beam Angle(50%) C0-180,C90-270: 84.1 deg,84.0 deg

C0-180 — C90-270

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

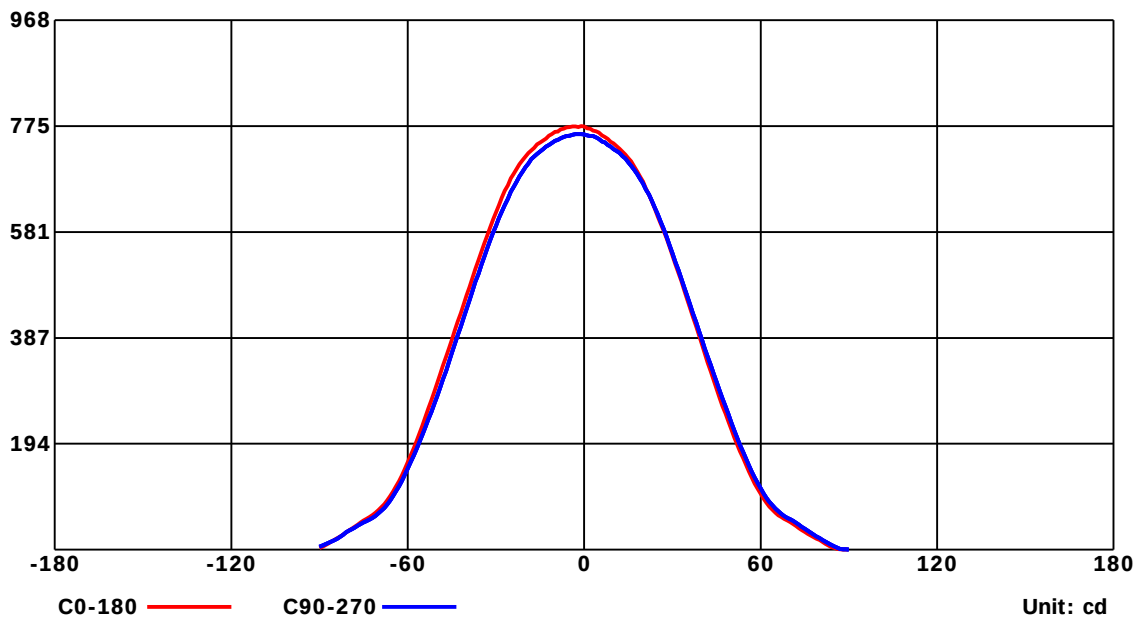
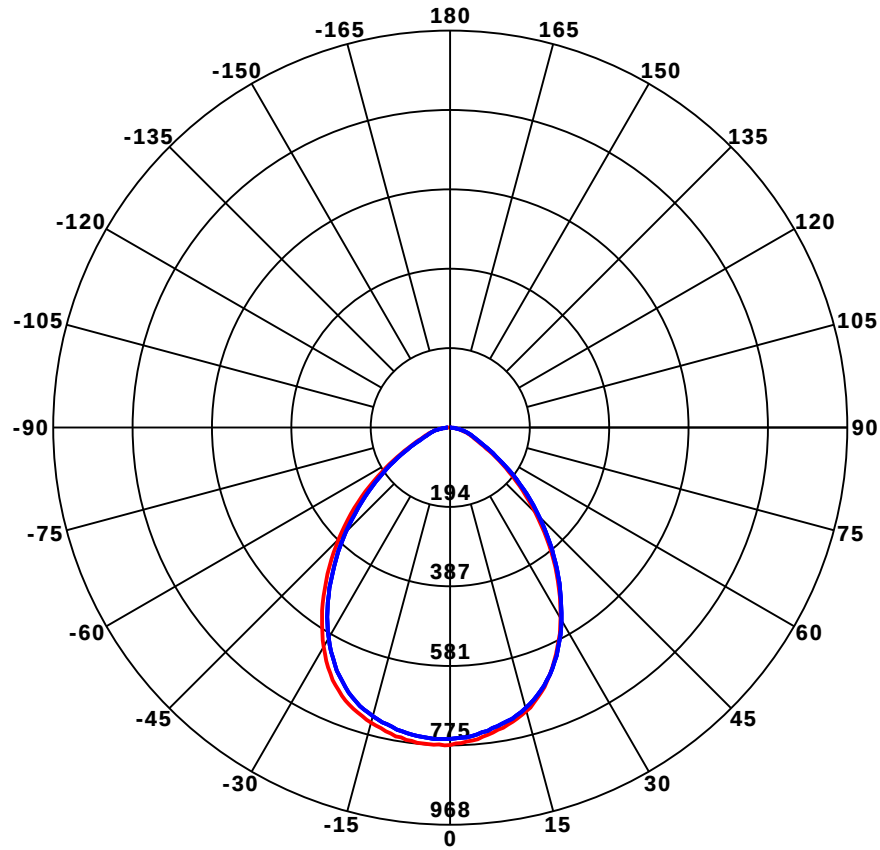
Test Distance: 6.25 m

Remarks:

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Luminous Intensity Distribution Curve



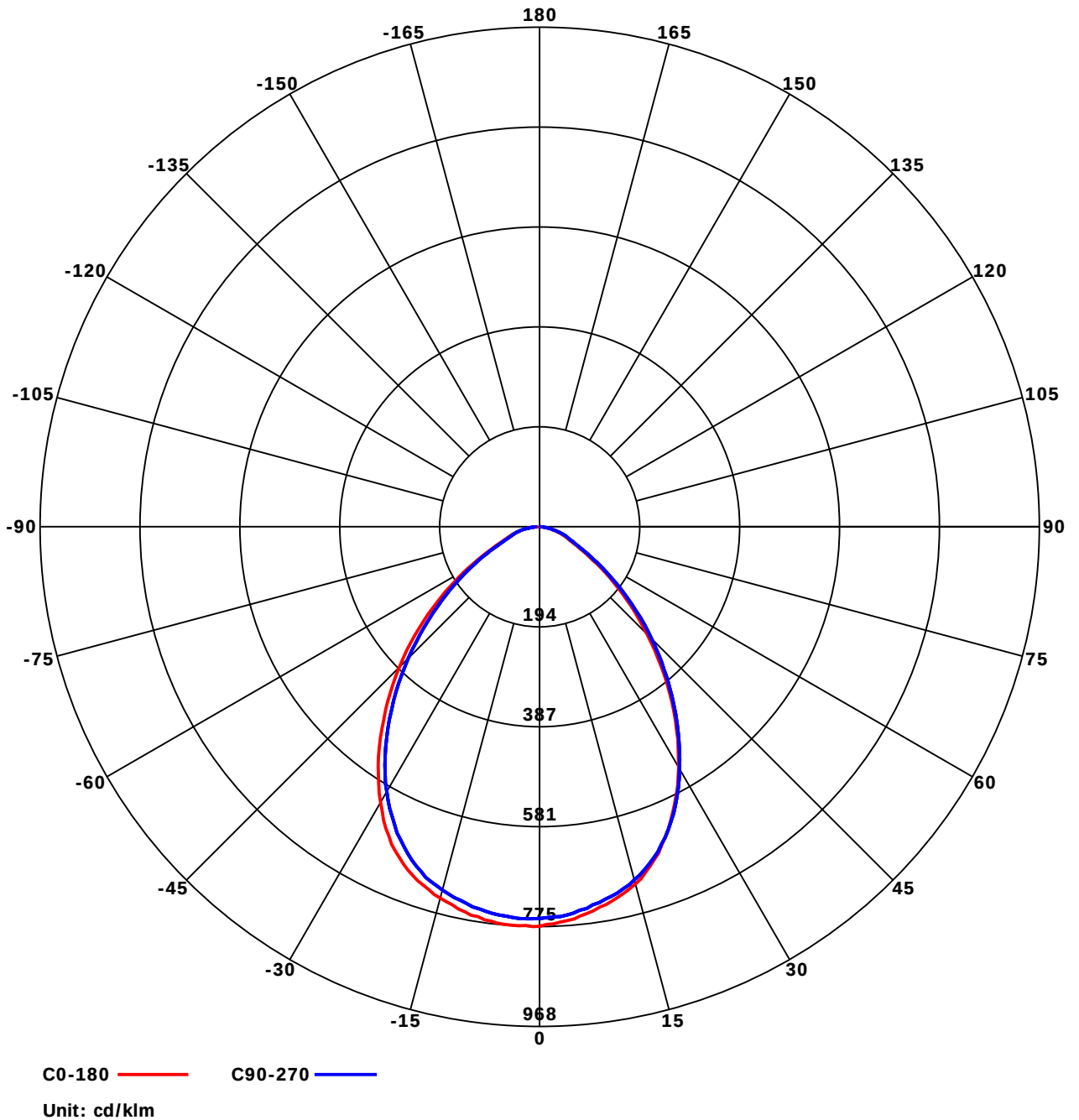
C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

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Luminous Intensity Distribution Curve



C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

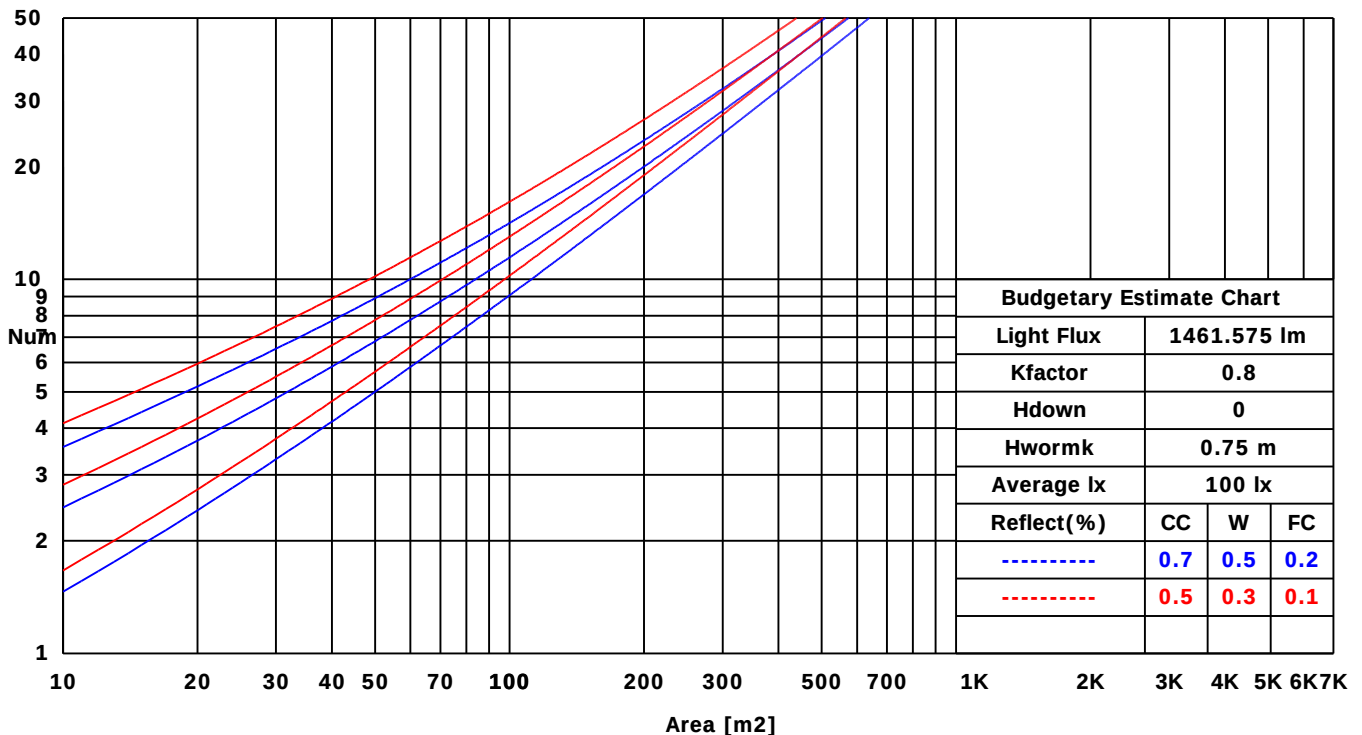
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UTILIZATION FACTORS TABLE

RHOCC	80			70			50			30			10			0
RHOW	50	30	10	50	30	10	50	30	10	50	30	10	50	30	10	0
RCR	COEFFICIENTS OF UTILIZATION FOR RHOFC=20															
0	1.18	1.18	1.18	1.15	1.15	1.15	1.10	1.10	1.10	1.05	1.05	1.05	1.01	1.01	1.01	0.99
1	1.08	1.07	1.06	1.07	1.05	1.04	1.03	1.01	0.99	0.98	0.96	0.94	0.91	0.89	0.87	0.82
2	0.95	0.93	0.92	0.94	0.92	0.90	0.91	0.89	0.86	0.88	0.84	0.82	0.83	0.79	0.76	0.72
3	0.84	0.82	0.80	0.84	0.81	0.79	0.82	0.78	0.75	0.79	0.75	0.72	0.75	0.71	0.67	0.63
4	0.75	0.72	0.71	0.75	0.72	0.69	0.73	0.69	0.66	0.71	0.67	0.63	0.69	0.64	0.59	0.55
5	0.67	0.64	0.63	0.67	0.64	0.62	0.66	0.62	0.59	0.65	0.60	0.56	0.63	0.57	0.53	0.49
6	0.60	0.58	0.56	0.60	0.57	0.55	0.60	0.56	0.53	0.59	0.54	0.50	0.58	0.52	0.48	0.44
7	0.54	0.52	0.51	0.55	0.52	0.50	0.55	0.51	0.48	0.55	0.49	0.45	0.54	0.48	0.43	0.40
8	0.50	0.47	0.46	0.50	0.47	0.45	0.51	0.46	0.43	0.50	0.45	0.41	0.50	0.44	0.39	0.36
9	0.45	0.43	0.42	0.46	0.43	0.41	0.47	0.42	0.39	0.47	0.41	0.38	0.46	0.40	0.36	0.33
10	0.42	0.40	0.38	0.42	0.39	0.38	0.43	0.39	0.36	0.43	0.38	0.35	0.43	0.37	0.33	0.30

LUMINAIRE BUDGETARY ESTIMATE DIAGRAM



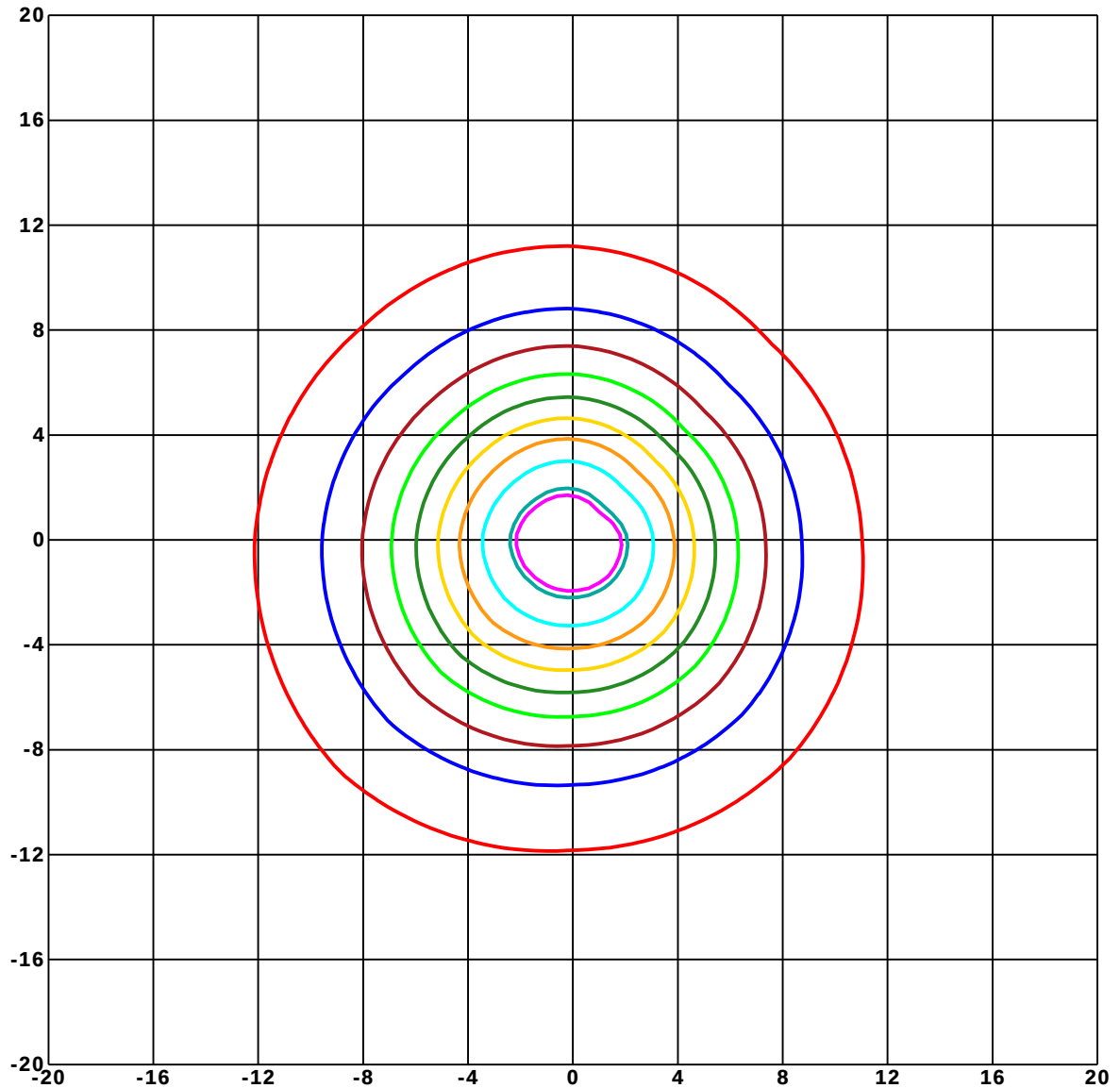
C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

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ISOLX DIAGRAM



Unit of illumination:lx Height of plane:10.0(m)/32.81(ft)

7.124(92%Emax)	3.871(50%Emax)
6.969(90%Emax)	3.097(40%Emax)
6.194(80%Emax)	2.323(30%Emax)
5.42(70%Emax)	1.549(20%Emax)
4.646(60%Emax)	0.774(10%Emax)

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

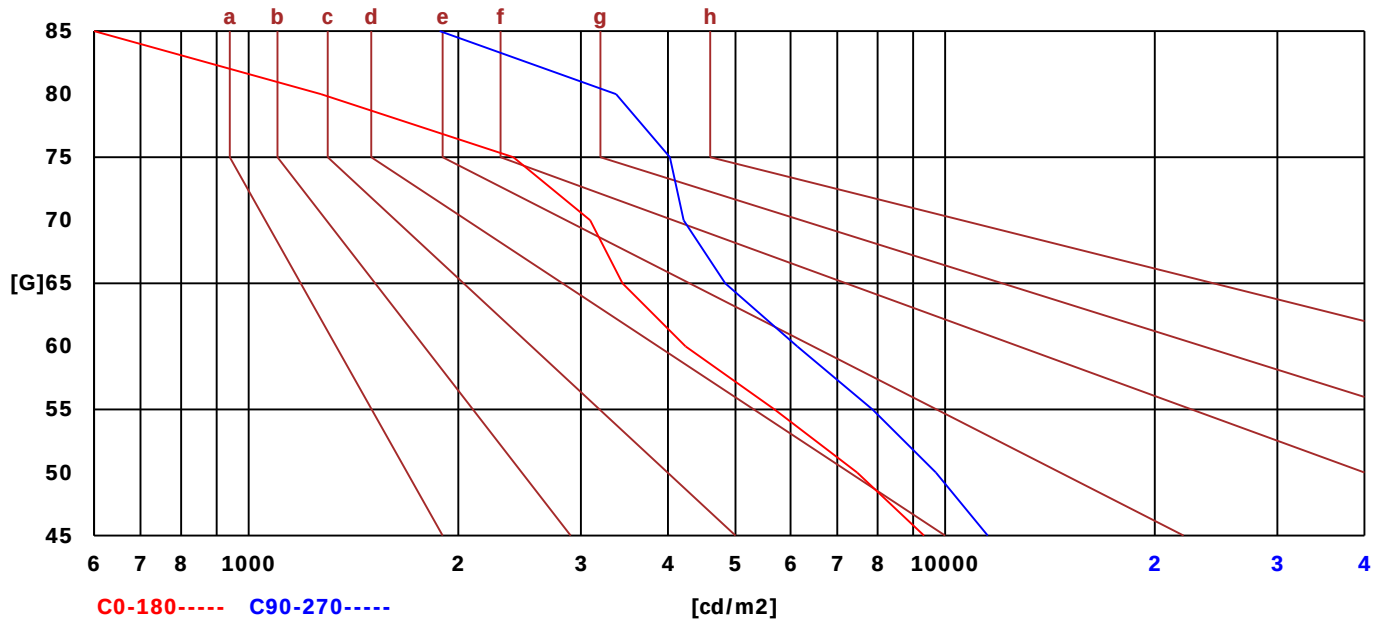
Remarks:

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BRIGHT LIMIT DIAGRAM

C/G	45	50	55	60	65	70	75	80	85
C0	9329	7478	5690	4242	3441	3095	2400	1271	0
C90	11515	9699	7864	6131	4834	4214	4026	3369	1882
Dazzle	Quality	Illuminance	---	---	---	---	---	---	---
1.15	A	2000	1000	500	<=300				
1.5	B		2000	1000	500	<=300			
1.85	C			2000	1000	500	<=300		
2.2	D				2000	1000	500	<=300	
2.55	E					2000	1000	500	<=300
---	---	a	b	c	d	e	f	g	h



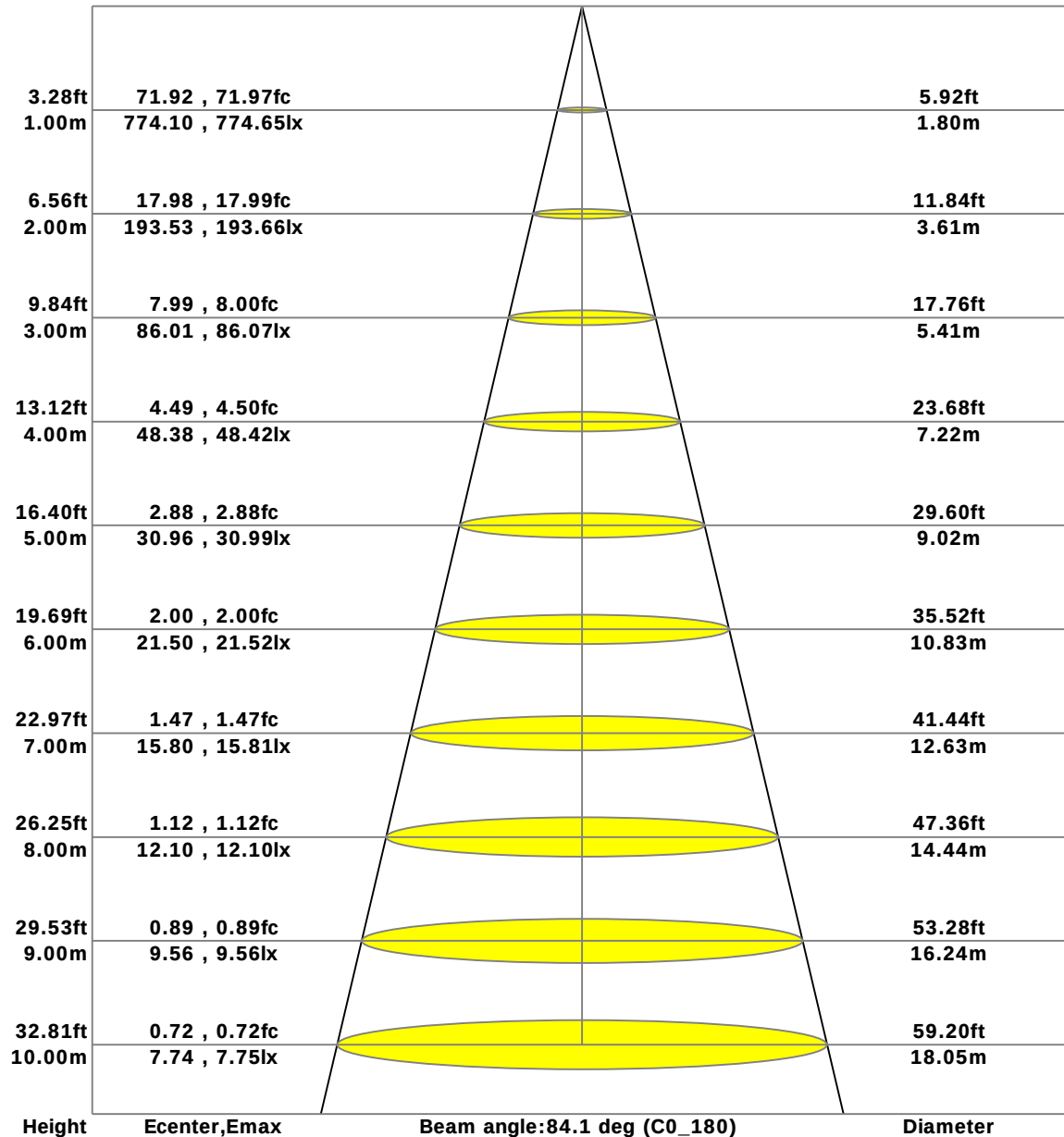
C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

Report No.:

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LX-DISTANCE DIAGRAM



C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

Report No.:

Test Time: 2024/10/15 11:37:18

UGR(Unified Glare Rating) TABLE

RHOCC	70	70	50	50	30	70	70	50	50	30
RHOW	50	30	50	30	30	50	30	50	30	30
RHOF	20	20	20	20	20	20	20	20	20	20
Room Size X Y	Viewed Endwise					Viewed Crosswise				
2H 2H	18.08	18.65	18.00	18.83	19.08	18.25	18.82	18.18	19.00	19.25
3H	17.76	18.29	17.75	18.51	18.80	17.93	18.45	17.92	18.67	18.96
4H	17.58	18.08	17.60	18.31	18.63	17.74	18.24	17.76	18.48	18.80
6H	17.38	17.85	17.44	18.10	18.45	17.54	18.01	17.60	18.26	18.61
8H	17.27	17.72	17.35	17.99	18.36	17.43	17.89	17.51	18.15	18.52
12H	17.16	17.60	17.27	17.88	18.26	17.32	17.76	17.43	18.04	18.42
4H 2H	18.63	19.13	18.65	19.36	19.68	18.91	19.41	18.94	19.65	19.97
3H	18.35	18.79	18.45	19.07	19.45	18.63	19.07	18.73	19.35	19.73
4H	18.17	18.57	18.31	18.87	19.28	18.44	18.84	18.59	19.14	19.56
6H	17.96	18.32	18.15	18.64	19.10	18.23	18.59	18.43	18.91	19.37
8H	17.85	18.18	18.07	18.52	18.99	18.12	18.46	18.34	18.79	19.27
12H	17.74	18.05	17.98	18.40	18.89	18.01	18.32	18.25	18.67	19.16
8H 4H	18.57	19.02	18.65	19.29	19.66	18.93	19.39	19.01	19.65	20.02
6H	17.93	18.21	18.20	18.58	19.09	18.29	18.57	18.56	18.93	19.45
8H	17.82	18.07	18.12	18.45	18.99	18.18	18.43	18.48	18.81	19.34
12H	17.71	17.93	18.04	18.32	18.88	18.07	18.29	18.39	18.68	19.23
12H 4H	18.04	18.35	18.29	18.70	19.20	18.43	18.74	18.68	19.09	19.59
6H	17.84	18.09	18.14	18.47	19.00	18.23	18.48	18.53	18.85	19.39

C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

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ZONAL LUMINOUS FLUX TABLE-1

[illegible]

C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

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ZONAL LUMINOUS FLUX TABLE-2

γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	φ zonal	φ total	%lum	%lamp
10	743.0	717.1	732.8	742.8	764.4	751.5	747.8	746.5	10	71.8	71.8	4.9	4.9
20	673.1	637.1	669.6	686.9	719.1	717.8	699.6	698.0	20	203.0	274.8	18.8	18.8
30	538.2	498.0	542.0	564.7	617.0	638.4	592.4	588.4	30	292.3	567.1	38.8	38.8
40	376.1	338.9	385.0	405.1	465.3	499.3	440.5	435.2	40	310.1	877.2	60.0	60.0
50	220.4	192.3	232.9	249.4	304.8	337.3	280.5	273.9	50	260.5	1137.7	77.8	77.8
60	101.9	84.8	111.9	122.6	159.9	193.4	148.2	141.6	60	172.6	1310.3	89.7	89.7
70	50.4	42.3	55.7	57.7	72.0	87.4	66.2	65.1	70	91.0	1401.4	95.9	95.9
80	17.6	8.8	21.0	23.4	35.0	44.5	34.7	31.2	80	46.3	1447.6	99.0	99.0
90	0.5	0.0	0.0	0.0	2.9	13.1	5.6	3.5	90	14.0	1461.6	100.0	100.0

Conical surface Flux(90deg) : 1016.9 lm

%lum = 69.6%

%lamp = 69.6%

Conical surface Flux(120deg) : 1310.3 lm

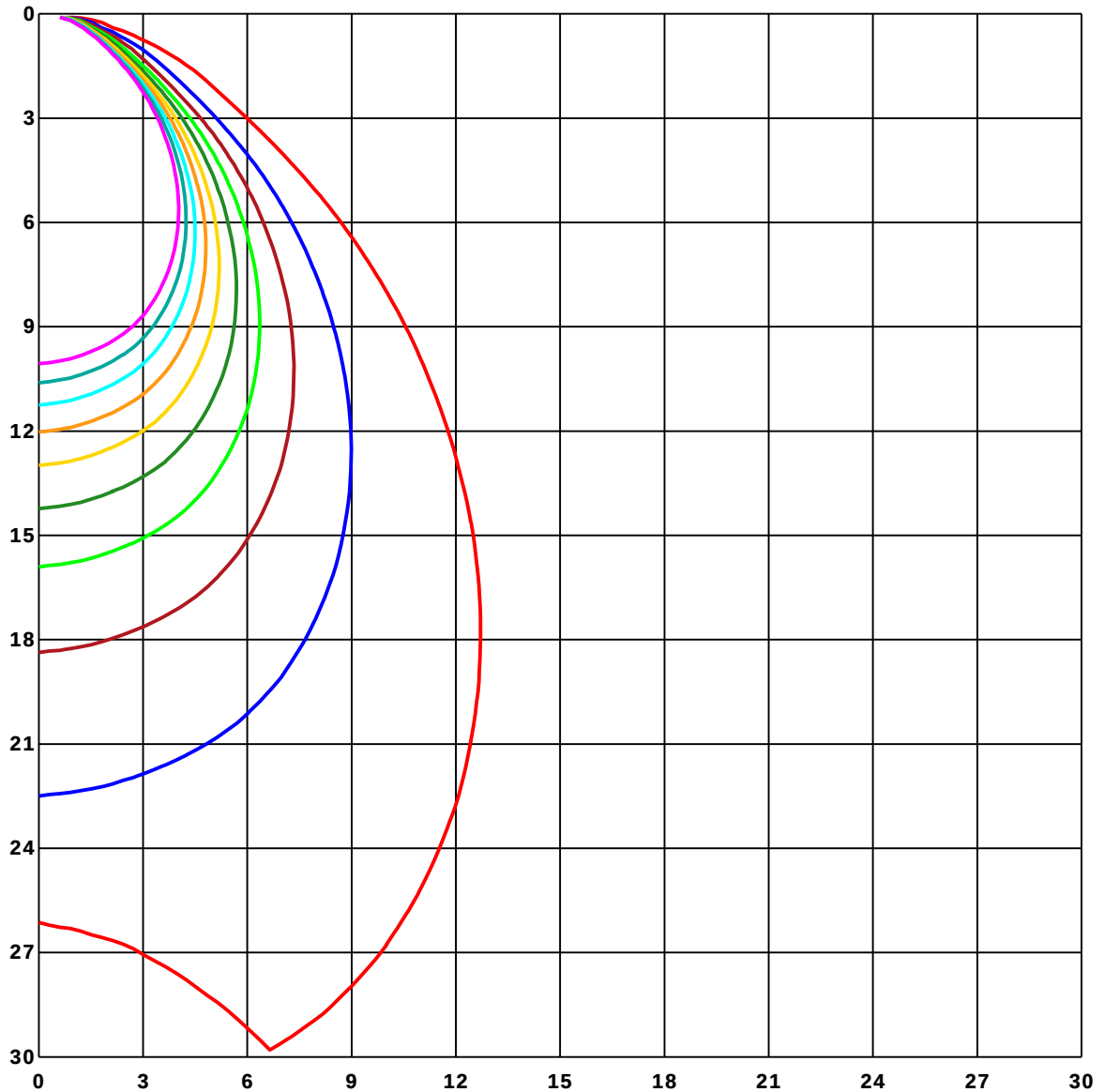
%lum = 89.7%

%lamp = 89.7%

Report No.:

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SPACE ISOLX DIAGRAM



Unit of illumination:lx

7.647
 6.883
 6.118
 5.353
 4.588

3.824
 3.059
 2.294
 1.529
 0.765

C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

Report No.:

Test Time: 2024/10/15 11:37:18

LUMINOUS DISTRIBUTION INTENSITY DATA

Gamma/C	C0	C45	C90	C135	C180	C225	C270	C315	C360	
G0	774.1	753.28	759.26	765.55	774.1	753.28	759.26	765.55	774.1	
G1	770.59	750.66	757.77	764.1	774.65	754.92	759.77	764.88	770.59	
G2	769.96	747.62	756.41	763.32	772.93	756.72	759.92	766.13	769.96	
G3	766.6	747.38	756.25	762.7	774.18	755.98	760.12	764.02	766.6	
G4	765.12	742.54	754.1	760.55	773.91	756.99	758.91	764.3	765.12	
G5	762.81	738.83	751.48	758.48	773.16	757.03	757.27	760.47	762.81	
G6	757.89	733.91	746.88	756.05	772.58	757.3	756.37	757.89	757.89	
G7	754.84	730.47	745.08	753.2	769.92	756.02	754.57	756.56	754.84	
G8	751.02	728.28	739.92	749.96	769.02	755.43	752.38	752.58	751.02	
G9	746.05	721.29	737.23	746.95	765.0	753.4	749.73	750.43	746.05	
G10	742.97	717.07	732.77	742.77	764.38	751.52	747.81	746.52	742.97	
G11	738.01	711.52	729.1	739.96	760.16	748.75	743.44	743.01	738.01	
G12	733.2	706.05	725.66	735.35	757.11	747.77	739.88	739.3	733.2	
G13	728.09	702.38	719.84	730.98	751.84	743.16	737.23	735.9	728.09	
G14	722.5	692.77	715.27	727.3	748.52	740.39	732.73	730.74	722.5	
G15	716.09	685.43	708.44	720.7	744.96	736.45	728.52	726.33	716.09	
G16	710.16	676.52	702.19	715.27	741.17	734.06	723.48	722.19	710.16	
G17	700.43	668.09	694.37	708.91	735.08	731.02	719.06	717.11	700.43	
G18	691.8	658.95	686.91	701.68	730.27	725.78	714.14	711.33	691.8	
G19	682.15	648.32	678.32	694.84	725.55	722.42	706.41	704.45	682.15	
G20	673.13	637.11	669.65	686.88	719.06	717.81	699.57	698.05	673.13	
G21	660.31	626.05	658.87	676.25	712.62	713.32	691.68	691.84	660.31	
G22	648.63	613.55	649.38	666.37	704.88	708.13	682.89	681.6	648.63	
G23	636.76	602.11	637.85	656.88	696.37	703.28	673.05	672.42	636.76	
G24	623.55	588.28	625.66	644.69	687.7	694.57	663.59	661.76	623.55	
G25	609.88	573.48	613.59	633.09	679.18	686.64	654.53	651.05	609.88	
G26	596.52	558.87	600.04	620.39	666.48	678.48	642.03	639.18	596.52	
G27	582.42	544.18	586.33	607.66	656.48	670.2	629.96	627.19	582.42	
G28	569.45	530.94	572.27	593.83	644.06	659.02	618.63	614.69	569.45	
G29	553.75	514.18	557.19	579.06	630.0	649.53	606.41	602.07	553.75	
G30	538.2	498.05	542.03	564.69	617.03	638.4	592.42	588.4	538.2	
G31	522.34	481.91	526.72	550.47	604.14	626.72	580.04	574.38	522.34	
G32	506.17	466.25	511.13	533.83	588.48	613.55	565.12	559.92	506.17	
G33	491.29	451.6	495.7	518.2	574.14	600.74	550.7	545.43	491.29	
G34	474.34	434.61	479.88	502.77	559.57	587.93	535.31	530.31	474.34	
G35	458.2	418.24	464.22	486.64	544.38	573.79	520.35	514.92	458.2	
G36	442.07	402.38	448.52	470.0	529.1	560.51	503.98	498.75	442.07	

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:37:18

LUMINOUS DISTRIBUTION INTENSITY DATA(Continue 1)

Gamma/C	C0	C45	C90	C135	C180	C225	C270	C315	C360	
G37	425.78	385.94	432.15	453.48	513.01	546.45	489.38	485.74	425.78	
G38	408.67	371.09	416.45	438.09	496.76	530.43	472.81	468.67	408.67	
G39	392.15	354.45	400.12	421.13	480.35	515.04	456.29	450.7	392.15	
G40	376.09	338.87	385.0	405.08	465.31	499.34	440.47	435.23	376.09	
G41	359.02	323.36	367.93	388.91	449.3	483.59	424.57	417.7	359.02	
G42	342.15	308.2	353.2	373.09	433.05	467.46	408.28	401.25	342.15	
G43	325.9	293.09	337.5	357.3	417.03	450.63	391.88	384.53	325.9	
G44	310.12	278.36	321.68	341.88	401.05	434.3	375.82	368.16	310.12	
G45	294.1	263.87	306.88	325.7	384.38	417.81	360.27	352.11	294.1	
G46	278.44	248.91	291.56	310.16	368.4	401.29	343.63	335.66	278.44	
G47	263.16	234.69	276.88	295.04	353.01	385.55	327.11	319.73	263.16	
G48	248.79	220.51	261.95	280.27	337.07	369.57	311.8	304.22	248.79	
G49	234.57	206.05	247.54	264.45	320.63	352.85	295.86	288.91	234.57	
G50	220.35	192.27	232.85	249.38	304.8	337.27	280.55	273.95	220.35	
G51	205.86	178.98	218.59	235.04	289.02	322.19	266.68	259.38	205.86	
G52	192.66	166.13	205.23	220.59	273.79	306.29	251.99	244.77	192.66	
G53	179.69	153.87	191.64	206.68	258.01	290.86	238.01	230.16	179.69	
G54	167.34	141.76	178.55	193.24	242.38	276.21	224.06	216.41	167.34	
G55	155.04	130.55	166.41	180.43	227.46	261.52	210.43	202.66	155.04	
G56	143.2	119.92	154.26	167.7	213.44	247.27	197.54	189.53	143.2	
G57	131.8	110.04	142.3	155.7	199.02	233.52	184.73	176.84	131.8	
G58	120.82	100.86	131.6	144.49	185.59	219.69	172.27	164.73	120.82	
G59	110.74	92.3	121.37	133.16	172.5	206.6	159.8	152.85	110.74	
G60	101.88	84.84	111.88	122.62	159.92	193.44	148.16	141.56	101.88	
G61	93.13	78.24	102.7	112.85	147.42	180.82	137.23	130.86	93.13	
G62	85.51	72.23	94.45	103.95	135.94	168.67	126.25	120.74	85.51	
G63	78.87	67.27	87.19	95.59	125.39	156.29	116.17	111.02	78.87	
G64	72.85	62.5	80.86	88.28	115.39	144.84	106.8	102.27	72.85	
G65	67.7	58.16	75.23	81.72	106.13	133.52	98.01	94.1	67.7	
G66	63.09	54.45	70.16	75.78	97.77	122.97	89.96	86.8	63.09	
G67	59.61	51.52	65.55	70.39	90.35	113.05	82.77	80.39	59.61	
G68	56.29	49.06	61.56	65.78	83.44	103.55	76.52	74.53	56.29	
G69	53.16	45.7	58.32	61.41	77.19	95.0	71.13	69.45	53.16	
G70	50.35	42.34	55.7	57.66	71.99	87.42	66.21	65.08	50.35	
G71	46.76	38.95	52.66	54.77	67.42	80.78	61.84	61.17	46.76	
G72	43.2	34.96	49.18	51.8	63.55	74.88	58.05	57.93	43.2	
G73	39.53	31.33	45.66	48.44	59.8	69.57	54.96	55.12	39.53	

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:37:18

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Test Time: 2024/10/15 11:37:18

[illegible]

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks: